Saarthi Manual

Section 1: Project Overview and Architecture

◆ 1. INTRODUCTION

Saarthi is an AI-powered **smart assistive stick** designed specifically to help **visually impaired individuals** move around safely, independently, and confidently in dynamic environments. Unlike traditional white canes that can only detect obstacles on the ground, Saarthi detects **all types of obstacles**, including:

- Overhead and side obstacles
- Moving objects like bicycles, dogs, or vehicles
- Stairs (including *descending* ones — which many devices fail to handle)
- And all this, **even in rainy or harsh weather**

It combines **computer vision**, **machine learning**, **ultrasonic sensing**, and **audio feedback**, all built into a **lightweight, waterproof stick** and a **waist pouch** worn by the user.

◆ 2. OBJECTIVE OF SAARTHI

The goal of Saarthi is simple yet powerful:

"To build a reliable, low-cost, smart mobility assistant that empowers blind users to navigate the world with safety and dignity — no matter the weather, terrain, or crowd."

◆ 3. REAL-WORLD CHALLENGES SOLVED

Traditional Canes	Saarthi Smart Stick
Detect only at ground level	Detect at multiple levels (front, top, side, down)
Can’t detect moving objects	Real-time object detection using AI
No info about stairs	Specially designed for **stair-down** detection
Can’t be used in rain	**Waterproof design** allows outdoor use anytime
No feedback to user	**Audio alerts** tell the user what’s ahead

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◆ 4. TARGET USERS

Saarthi is designed for:

- Blind or low-vision individuals
- Especially those in urban and semi-urban areas
- Users who frequently move alone in public places like streets, campuses, offices, etc.

It can also be used in **rehabilitation centers**, **special schools**, or **research labs** that work on accessibility tools.

◆ 5. CORE CONCEPT

Saarthi combines:

- **Ultrasonic sensors** to measure distances to nearby obstacles and stairs.
- **Camera** to capture real-time video input.
- **AI object detection** model (MobileNet SSD v2, optimized with Google Coral USB).
- **Text-to-Speech (TTS)** to speak out alerts through a small speaker.
- **Power bank-based power system** to ensure portability.
- **Raspberry Pi 5** as the processing brain.

◆ 6. MAIN FEATURES OF SAARTHI

1. 🚶 Obstacle detection (3 directions)
2. 📷 AI-based object recognition
3. 🔊 Audio feedback using speaker
4. 💧 Fully waterproof and rugged
5. 🎒 Two-part architecture: lightweight stick + waist pouch
6. ☔ Works **even in rainy weather**
7. 🔻 **Can detect downward stairs** (biggest failure of most smart canes!)

◆ 7. OVERCOMING WEWALK'S DRAWBACKS

WeWALK Limitation	Saarthi Solution
✗ Cannot detect stairs going down	✓ Uses downward-angled ultrasonic sensor
✗ Not reliable in rainy conditions	✓ Waterproof tape, sealed circuits, power bank pouch

Saarthi was purposefully engineered to **fix these gaps** and show your true capability as a tech innovator.

◆ 8. SYSTEM DESIGN: STICK + WAIST POUCH

We break the system into two parts:

□ Stick (Handheld)

- Contains:
  - 1x front-facing ultrasonic sensor
  - 1x downward-facing ultrasonic sensor
  - 1x small speaker (near handle)
  - Wire routing toward the waist pouch
- Must be:
  - Lightweight (foldable aluminum)
  - Waterproof (sealed with insulation tape)

🎒 Waist Pouch (Worn on Belt or Waist)

- Contains:
  - Raspberry Pi 5 (inside plastic waterproof box)
  - Google Coral USB Accelerator (connected via USB 3.0)
  - Raspberry Pi Camera (mounted on pouch or chest belt)
  - Power bank (10,000 mAh)
- Why pouch?
  - To avoid overloading the stick and tiring the user
  - To protect electronics from external weather

◆ 9. BLOCK DIAGRAM (TEXTUAL FORM)

```
 [Camera + Ultrasonic Sensors]
            |
            v
[Raspberry Pi 5 + Coral USB]
            |
            v
[Object Detection + Obstacle Distance Logic]
            |
            v
[Text-to-Speech System]
            |
            v
[Speaker Audio Output]
```

Explanation:

1. Camera captures video frames.
2. Raspberry Pi + Coral USB runs MobileNet SSD v2 to detect objects.
3. Ultrasonic sensors measure distance.
4. If an object is close or in path, the Pi triggers a warning.
5. Audio alert (e.g., “Dog ahead”, “Stairs ahead”) is spoken via speaker.

◆ 10. HIGH-LEVEL WORKFLOW

Let’s break it step-by-step like a flowchart:

1. **User powers on the system** using power bank.
2. **Camera starts capturing the front view.**
3. **AI model detects known objects** (dog, person, car, cycle, etc.).
4. **Ultrasonic sensor measures proximity** (distance to object).
5. **If object is too close or it’s a stair:**
    - System generates a **voice alert**
6. **Speaker plays audio**, telling the user what’s coming.
7. **User responds accordingly** (slows down, turns, or avoids).

This system is highly practical, responsive, and built to be weatherproof and energy-efficient.

◆ 11. GOALS OF SAARTHI PROJECT

- ✅ Show that AI + IoT can change lives, not just impress teachers
- ✅ Build a **real**, not theoretical, solution
- ✅ Beat existing products with creativity + practicality
- ✅ Keep everything **cost-effective and reproducible**

□ FINAL TAKEAWAY FROM SECTION 1:

You are not just making a project — you are building a **proof of potential**, something that solves a **critical problem** smartly and shows that technology should serve **people who need it the most**.

The rest of the manual will walk you step-by-step through:

- Buying the right components
- Wiring them correctly
- Coding the software
- Testing everything safely
- Waterproofing and final demo-ready assembly